



BEAN AND BREW



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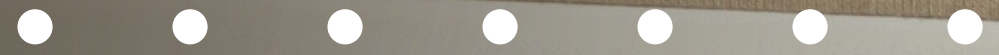
...

Brewing Sustainability

SIP, SAVOR, SMILE.

...

The Outliers





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OUR PROBLEM AND FOCUS

Sustainability & Profitability

Bean & Brew needs a **data-driven strategy** to increase reusable cup usage without hurting revenue or customer experience. After mixed results from the 2023 surcharge and rebrand, the café must understand **customer behaviour**, measure **financial impacts**, and implement a phased **sustainability plan** that is both effective and feasible.

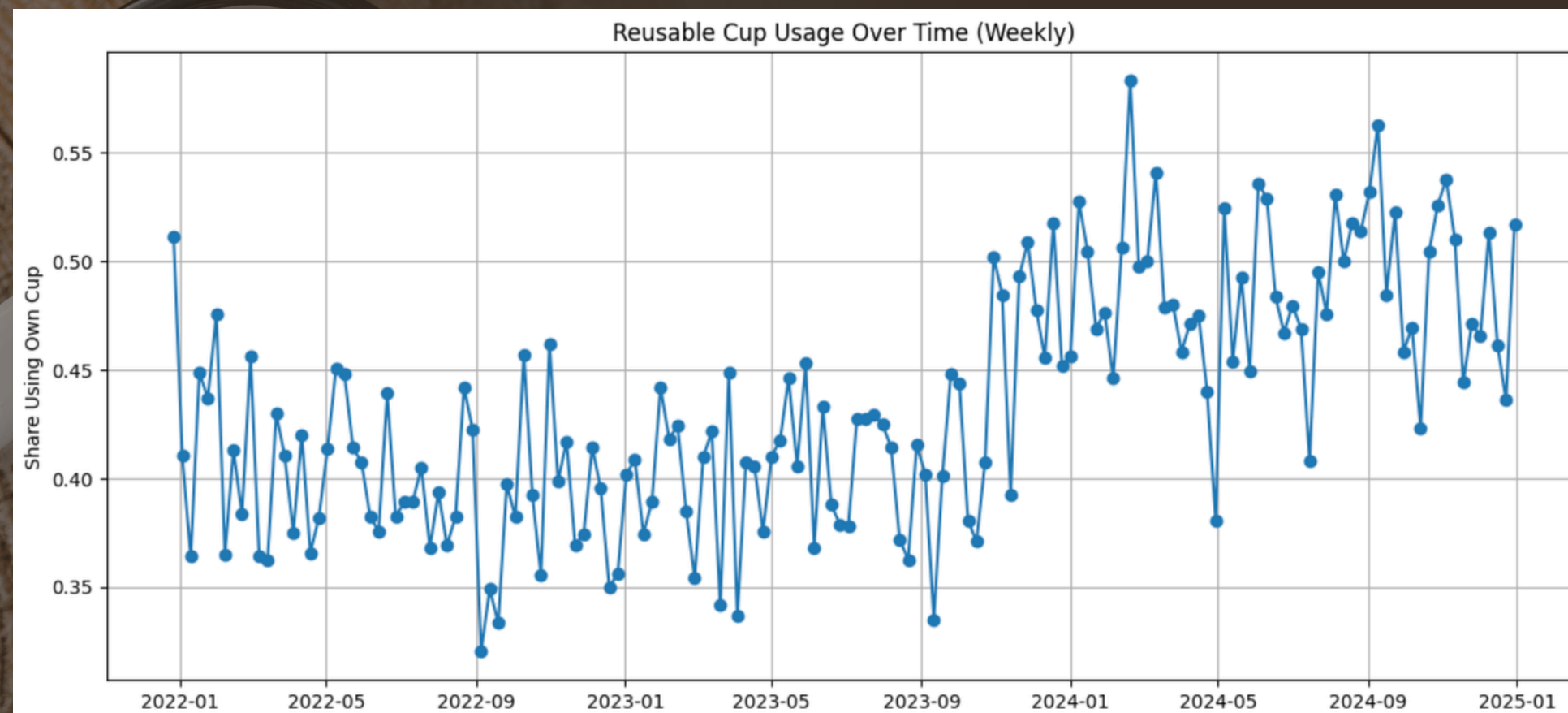
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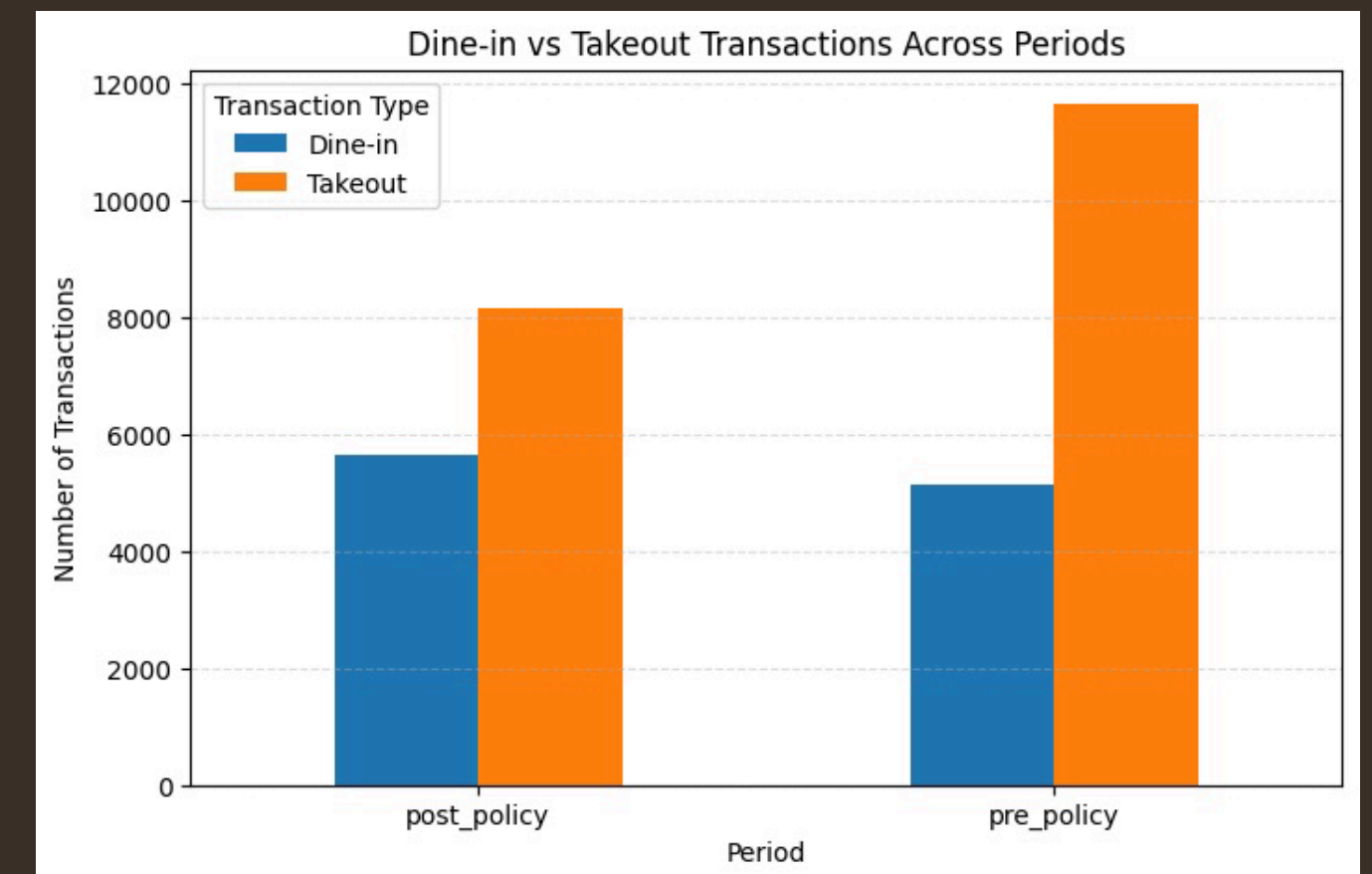


OUR ANALYSIS

In November 2023, Bean & Brew added a surcharge on disposable cups and encouraged customers to bring reusables or dine in. This was paired with a sustainability-focused rebrand to promote lower waste and a stronger study-friendly café vibe.



Reusable cup usage has steadily increased over the three-year period, rising from roughly the high-30% range to just above 50%.



Policy charge appears to have pushed customers toward dine-in, which increased, while takeout transactions fell sharply, showing a clear shift away from takeout after the charge

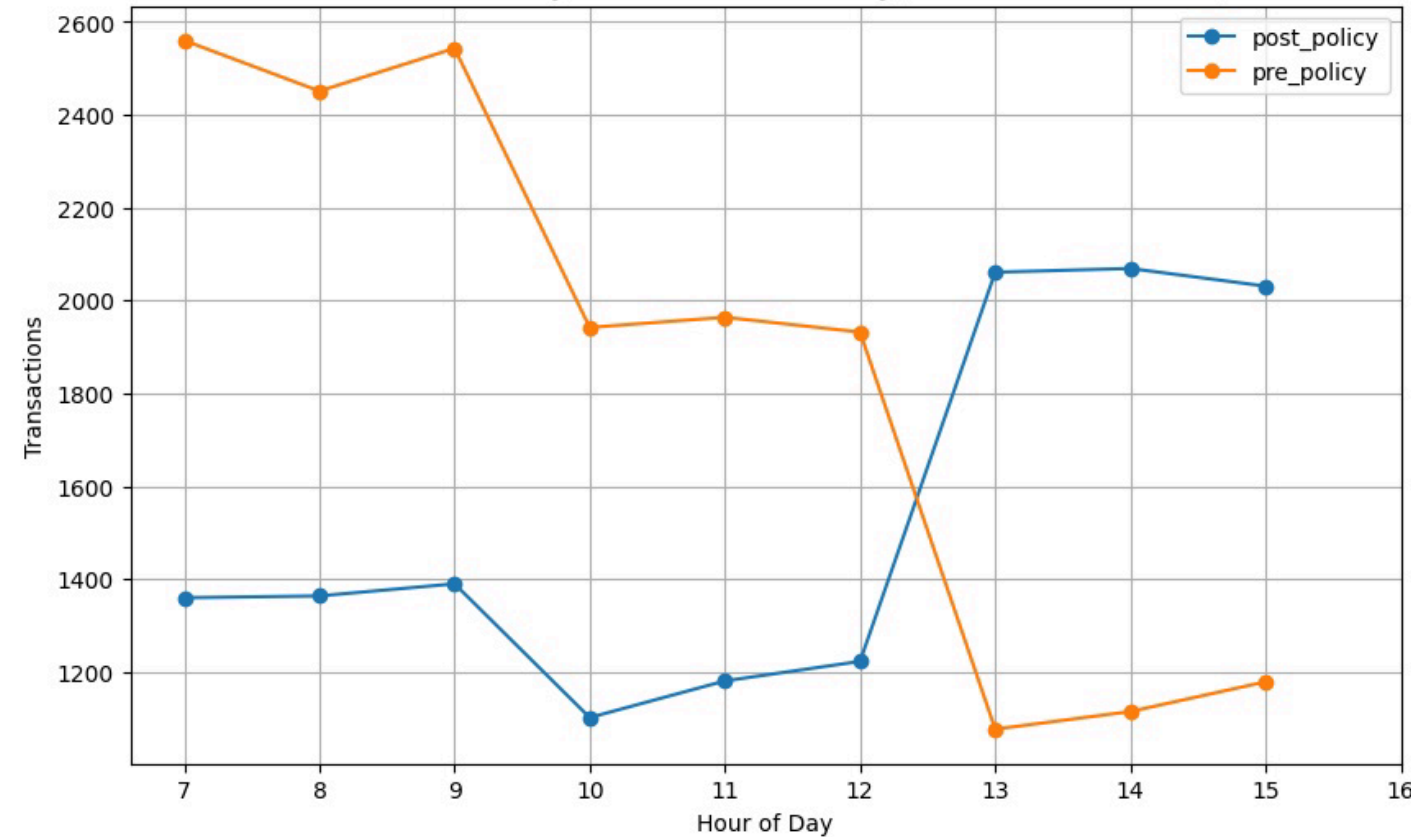


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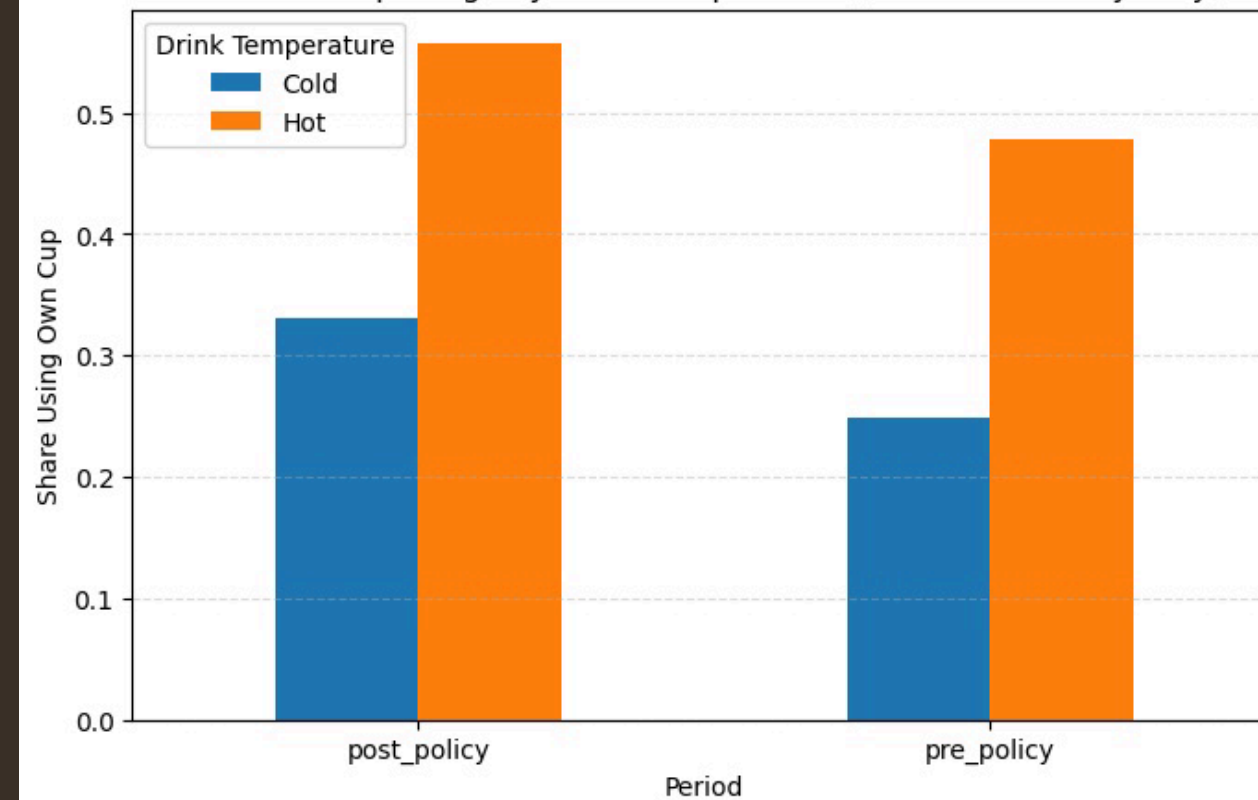
LIMITATIONS WITH CURRENT POLICY

Customer Traffic by Hour: Pre vs Post Policy (Construction Excluded)



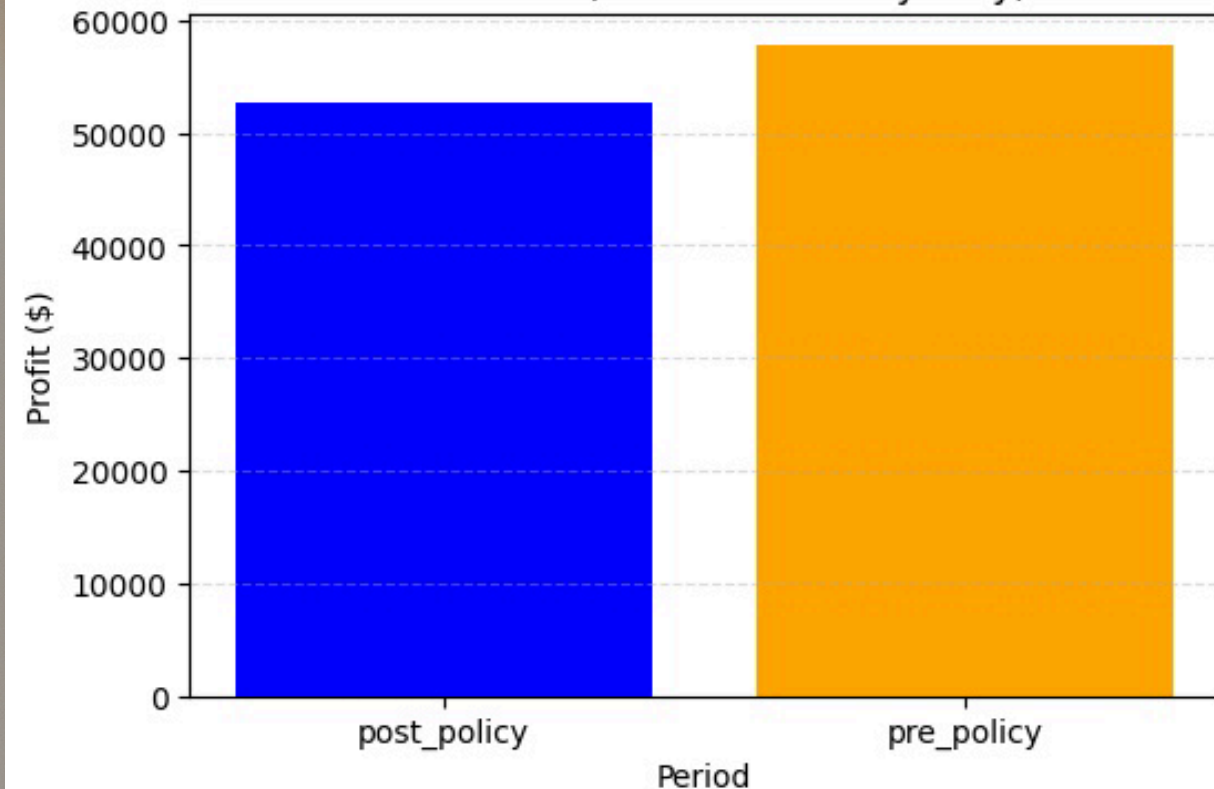
Post-policy morning traffic clearly drops compared to the pre-policy period.

Reusable Cup Usage by Drink Temperature (Pre vs Post Policy Only)



Reusable cup use rises overall, but cold drinks still lag behind hot drinks by a noticeable margin.

Total Profit (Pre vs Post Policy Only)



Post-policy profit decreases slightly compared to the pre-policy period.

Period	Revenue	Profit
Pre-Policy (before november 2023)	81,485.30	57,764.05
Post-Policy (after november 2023)	74,890.90	52,632.20



OUR RECOMMENDATIONS

The segment most affected by Bean & Brew’s sustainability shift is the weekday morning takeout commuter, whose routine is highly sensitive to pricing and convenience.



Phase 1

Start Simple (Jan–Mar)

- Morning & dine-in discounts
- Pilot deposit–refund
- Build baseline data
- Buy a merch - cup get 50% discount



Phase 2

Build Systems (Apr–Aug)

- Introduce check-in
- Sustainability dashboard
- Scale deposit–refund + credits



Phase 3

Full Engagement (Sep–Dec)

- Rewards + seasonal merch
- High-impact sustainable incentives
- Subscription based plan





Phase 1 *(Jan–March)*

GOAL: Test low-risk incentives
+ build baseline behaviour
before major policy changes.

Costs to Consumers per Cup	\$0.75
Subscription Costs to the Company per Cup	\$0.40

1. Happy Morning Hours

- Morning discount on drinks (10%)

Encourages early-day traffic — especially weekday regulars.

- Dine-in discount (20%)

Supports the “stay-and-study” vibe and reduces disposable cup usage naturally

2. Pilot Friendlier Subscription (Deposit–Refund Policy)

Introduce a soft-launch pilot:

- Introduce Friendlier’s reusable cup system for a small test period, customers borrow and return cups through platform
- Observe customer reaction and ease of adoption and evaluate operational fit before scaling up

3. Buy a Merch Cup - get a 50% discount on first drink



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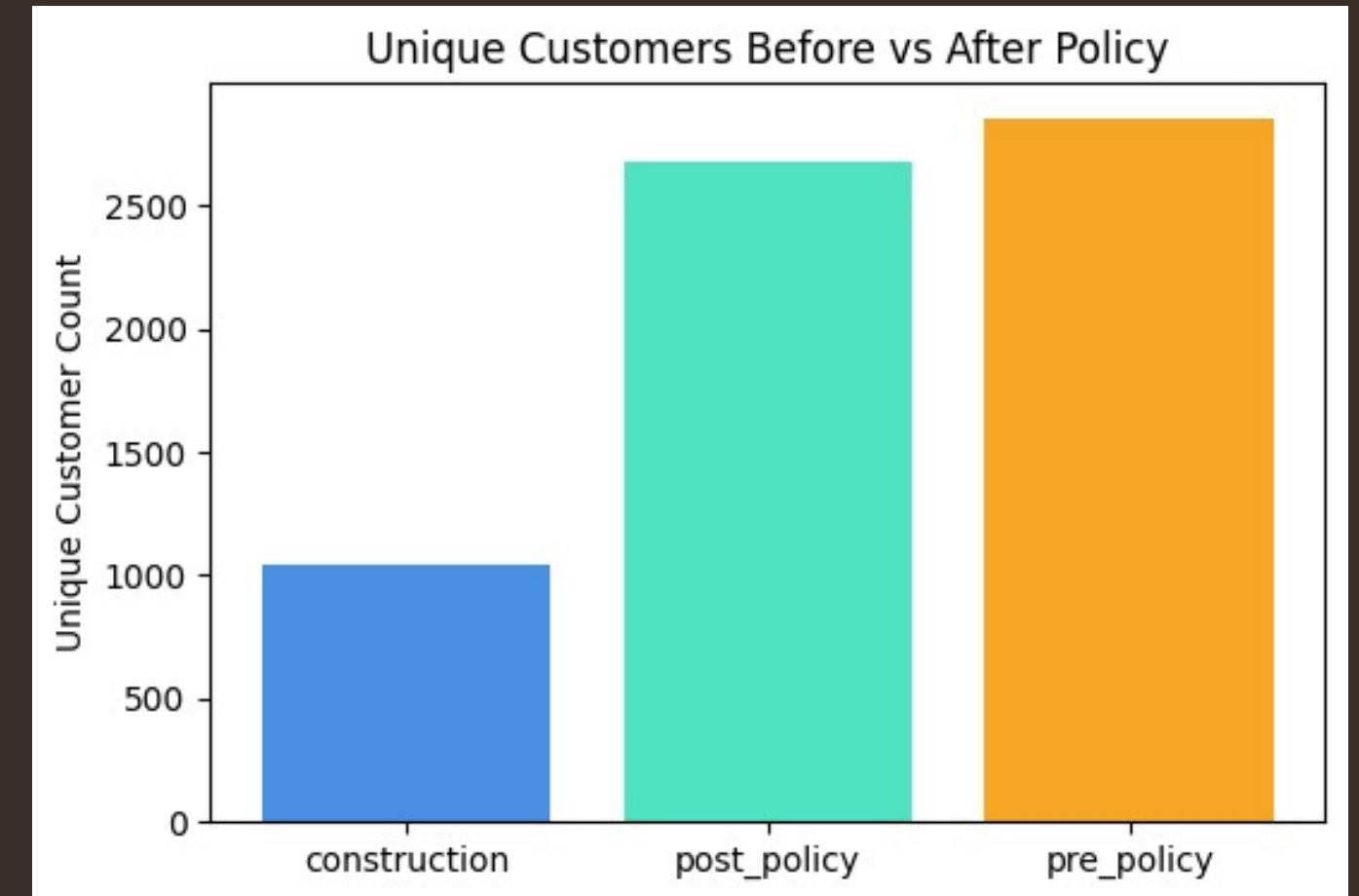
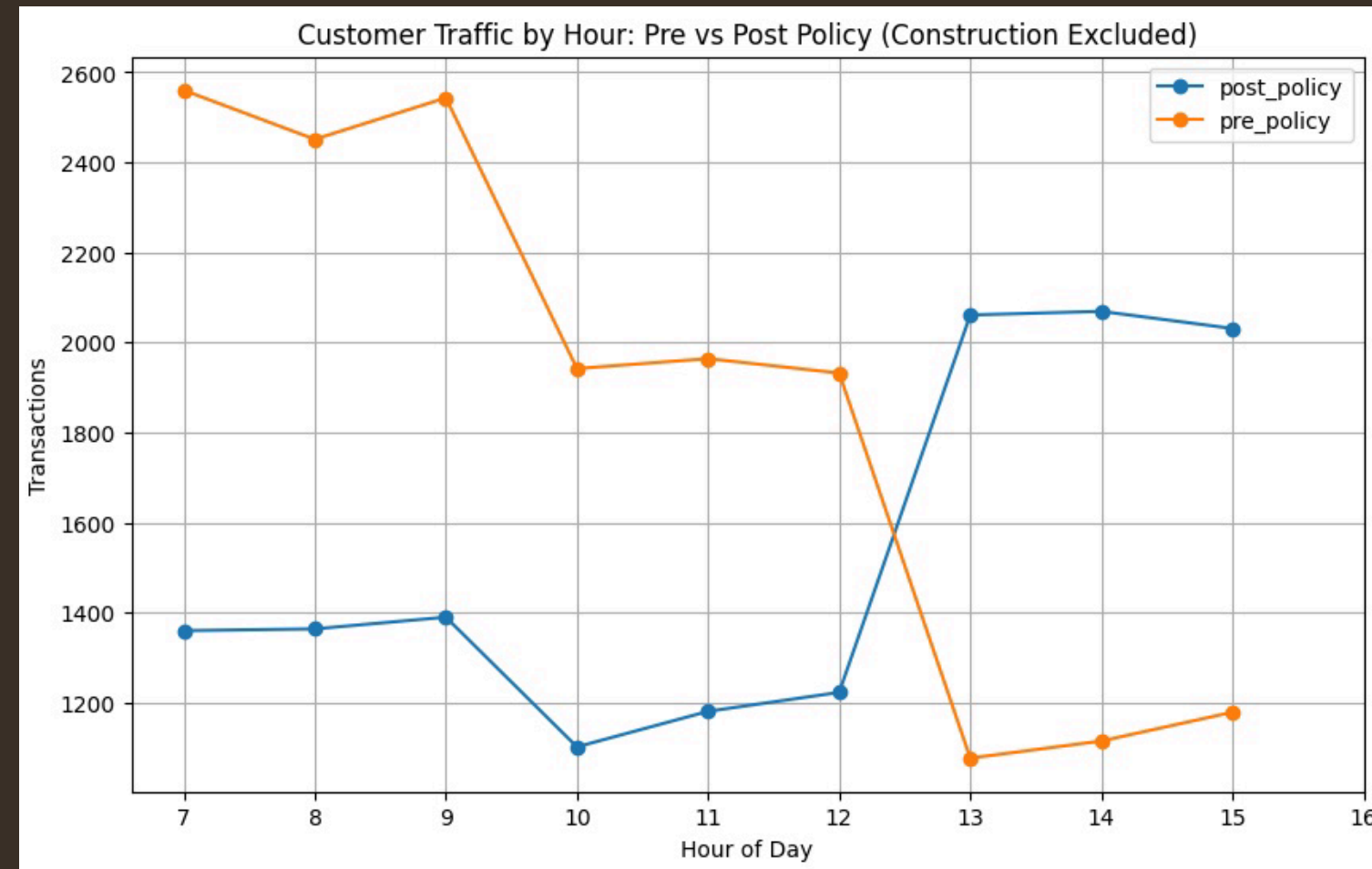
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Phase 1 (Jan–March)

WHY WE PICKED THOSE STRATEGIES



Z TEST (2 PROP): ===== 2-PROPORTION Z-TEST (PRE VS POST POLICY) =====
Z-STATISTIC: 14.434148210062817
P-VALUE: 3.1551955515979932E-47
PRE-POLICY REUSE: 0.4012647132481843
POST-POLICY REUSE: 0.486749680715198
SIGNIFICANT: POLICY CAUSED A REAL CHANGE IN BEHAVIOUR.
THIS SHOWED US THAT THE SHARE OF PPL USING REUSABLE CUPS IS GOING UP

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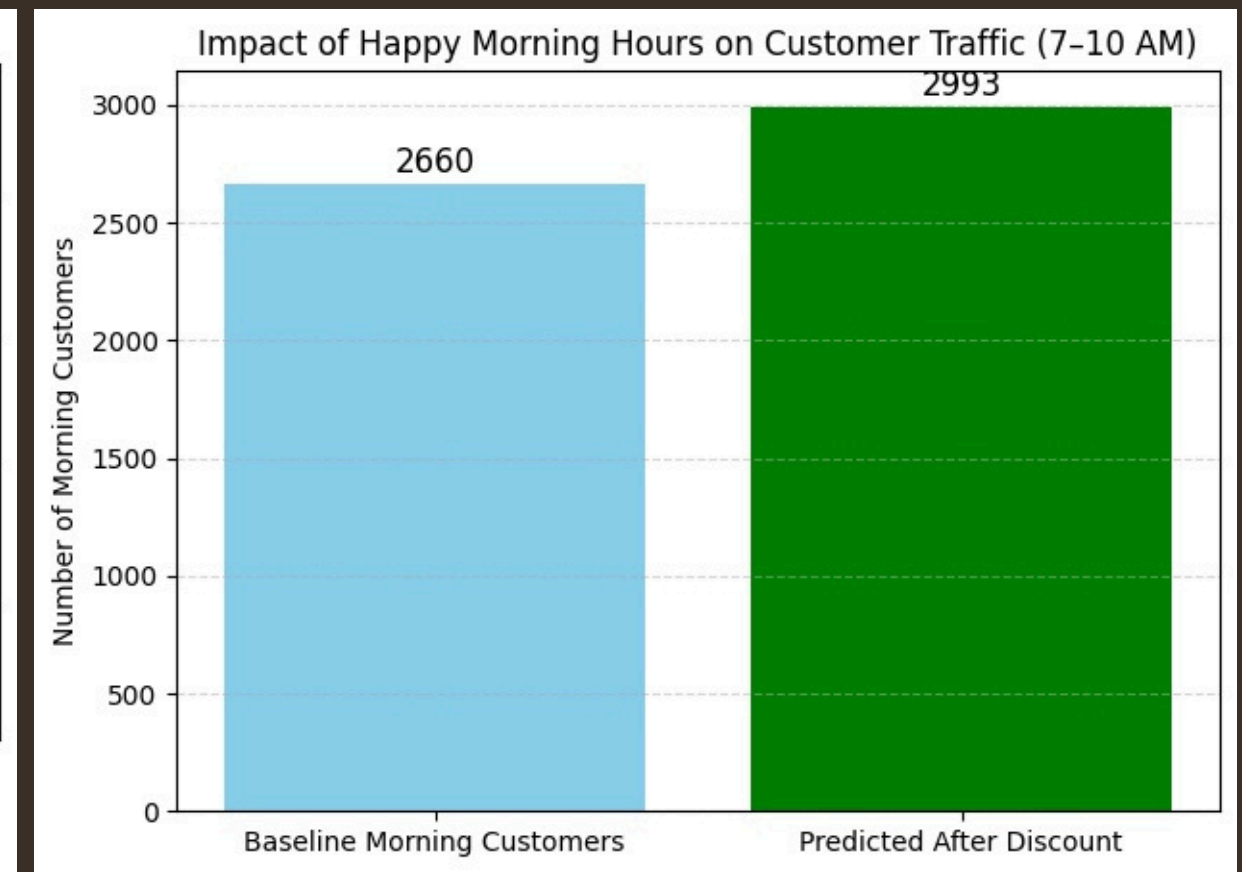
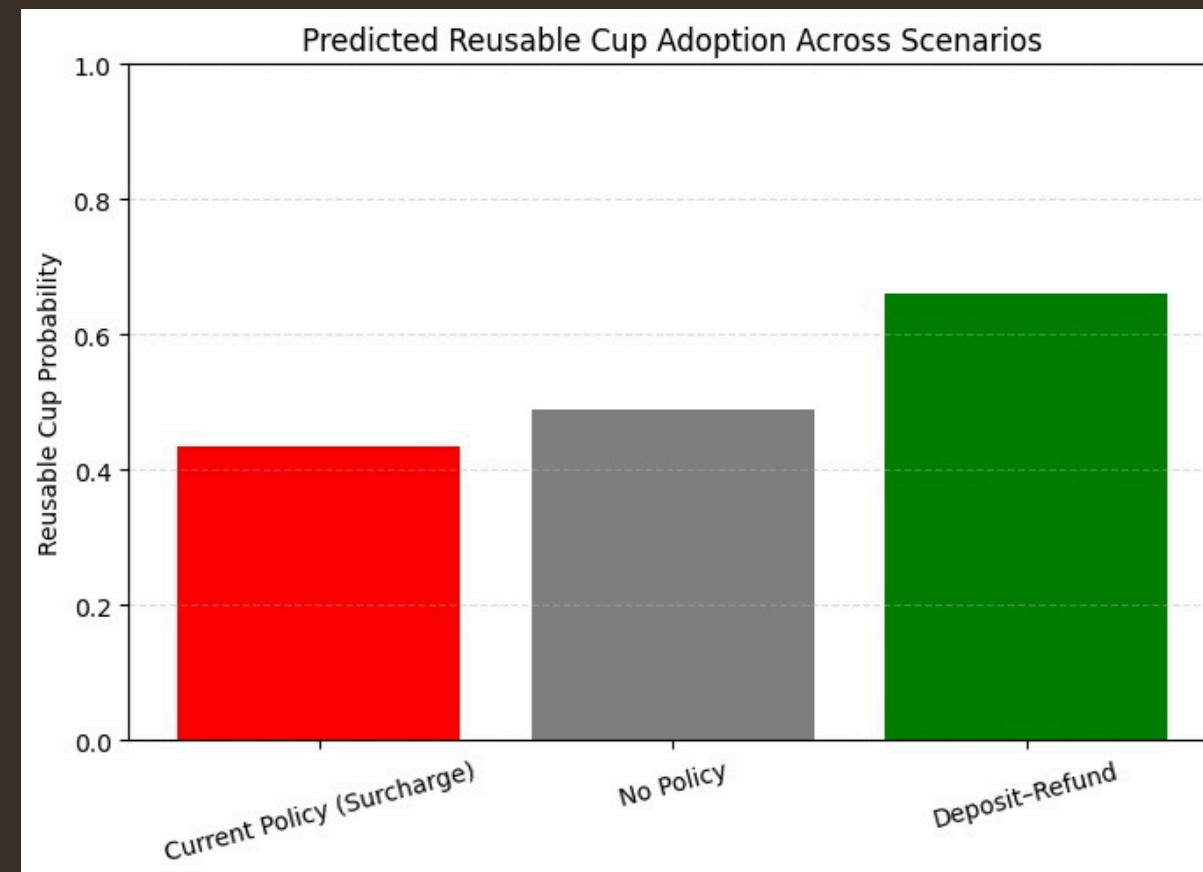
Phase 1 *(Jan–March)*

GOAL: Test low-risk incentives
+ build baseline behaviour
before major policy changes.

KPIs for Phase 1 (Key-Performance Indicator)

- Reusable cup adoption baseline
- Morning peak volume changes
- Net margin impact with discounts
- Dine-in duration + turnover

EFFECTS (PREDICTIONS FROM OUR MODEL):





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Phase 2 *(April-Aug)*

GOAL: Build structure, gather data, prepare for larger behavioural nudges.

KPIs for Phase 2

- Increase in reusable cup rate after dashboard
- Foot traffic change during check-in adoption
- Segmented behaviour insights (Morning vs Afternoon)
- Break-even point of deposit-refund system

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1. Measurement & Iteration System (Launch)

- Internal data at point-of-sale
- Tracks sustainability metrics, reusable cup performance, visit frequency, repeat behaviour, and campaign impact for ongoing optimization.

2. Sustainability Dashboard (Big addition)

Real-time In-Café Dashboard displaying:

- Cups saved this month
- % of customers using reusable cups
- Leaderboard for “Morning Regulars”, “Weekend Study Crew”, etc.

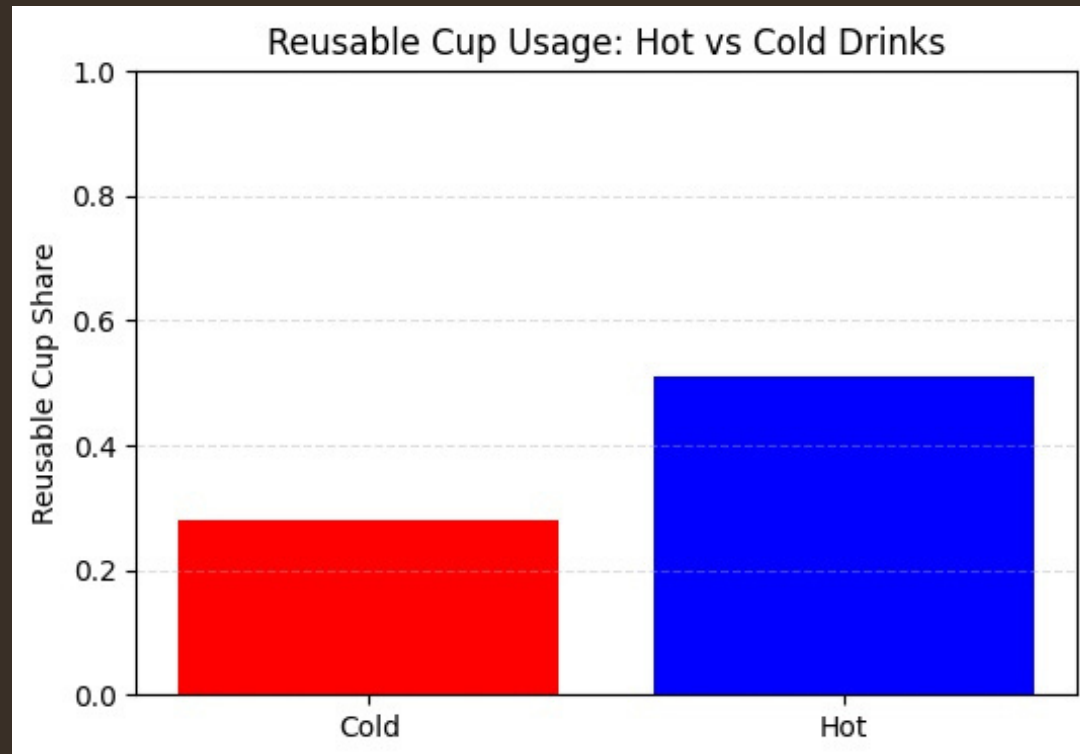
3. Deposit-Refund Scaling (if Phase 1 data supports it)

- Expand deposit-refund to introduce credit rewards for consistent returners
(e.g., “Return 10 cups → free drink credit”)



Phase 3 (Sep-Dec)

GOAL: Full-scale behavioural program aligned with school return rush.



1. Deposit-Refund Enhancements

- Credits → merch discounts
- Introduce seasonal reusable cups (Fall/Winter collection)

2. Offers for High-Engagement Users

Examples:

- Free drink after X sustainable check-ins
- Higher credit multipliers during peak weeks

3. Monthly Subscription-Based Plan

4. Dual Reusable Cup Policy (Hot vs Cold)

- Separate reusable cups by drink type for improved user experience
- Hot drinks: opaque, insulated cups for safety and heat retention
- Cold drinks: transparent, glass-style cups to satisfy visual preference



1. Revenue dilution: Discounts may lower short-term margins.
2. Operational load: Deposit–refund, check-ins, and returns adds training for staff.
3. Customer adoption risk: Some users may resist new systems or find them inconvenient.
4. Upfront setup costs: Dashboards require investment.
5. Behaviour dependency: Success relies on customers actually engaging.
6. Space constraints: Return bins, signage, and displays reduce limited café space.
7. Data dependency: We can only move into Phase 2 after confirming strong engagement from Phase 1.





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Thank You!



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APPENDIX

Dashboard

Beans and Brew – Interactive Analytics Dashboard

Use this dashboard in Q&A to answer judge questions live. Filter by date, choose a dimension for the x-axis, select a metric, and optionally focus on drinks only, loyal customers, periods, etc.

Date range: 01/01/2022 → 12/31/2024

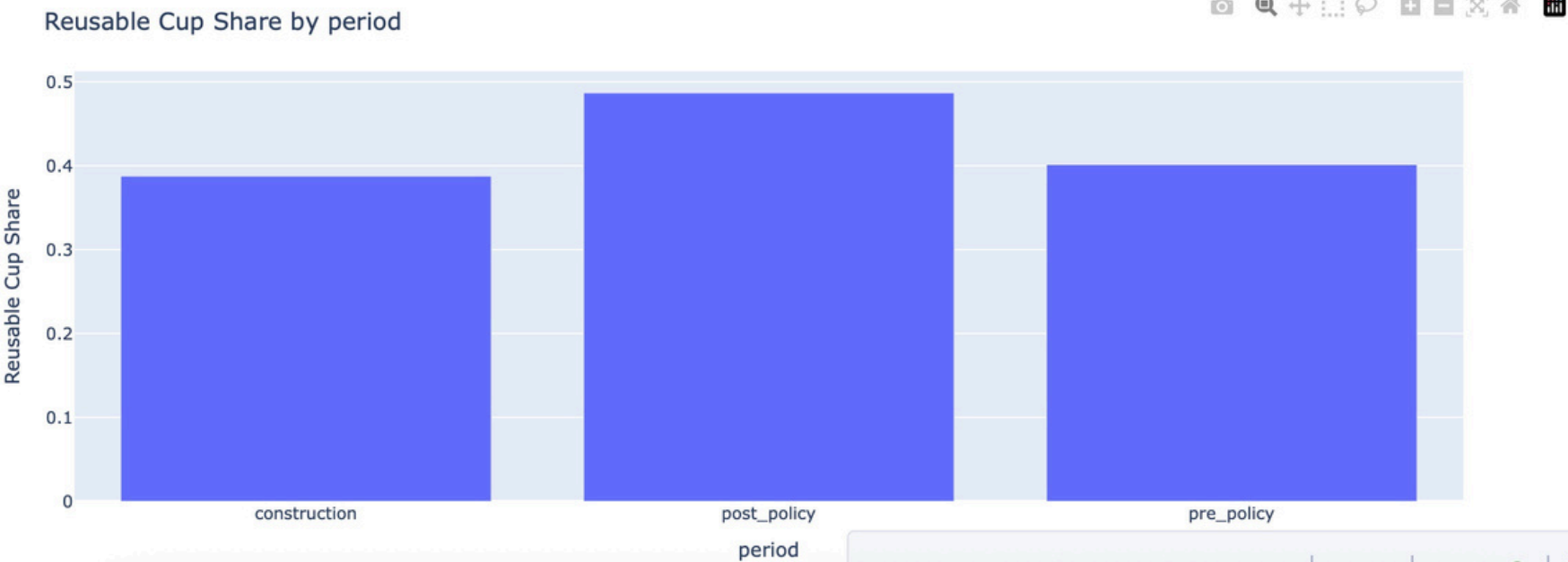
X-axis: Period (Pre / Constructio... ▼

Metric: Reusable Cup Share ▼

Scope: ☒ All Items ☐ Drinks Only

Periods
☒ Pre-policy ☒ Construction ☒ Post-policy

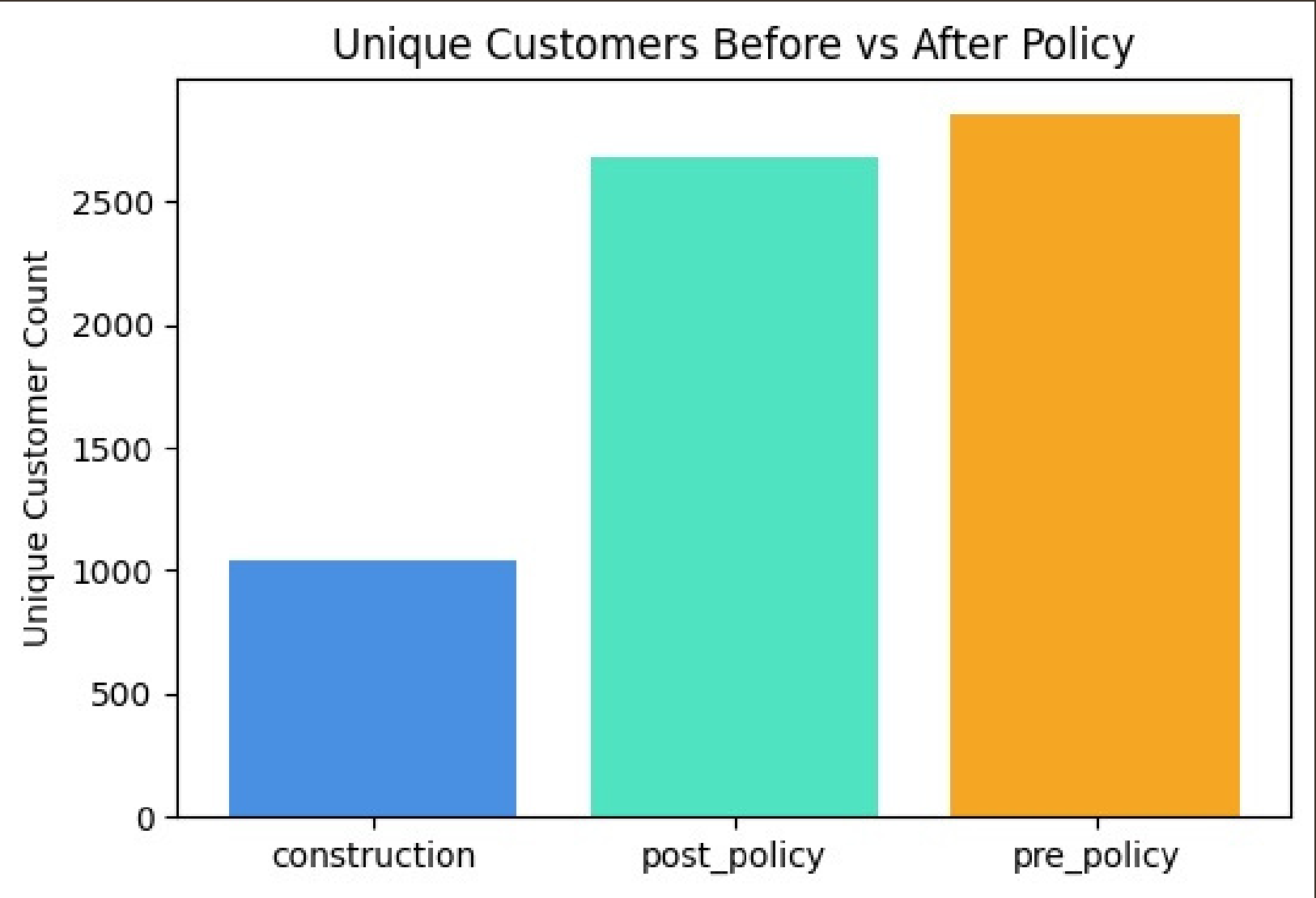
Total Transactions	Unique Customers	Avg Profit / Transaction	Reusable Cup Share (Drinks)
32,605	2,994	\$3.60	43.6%



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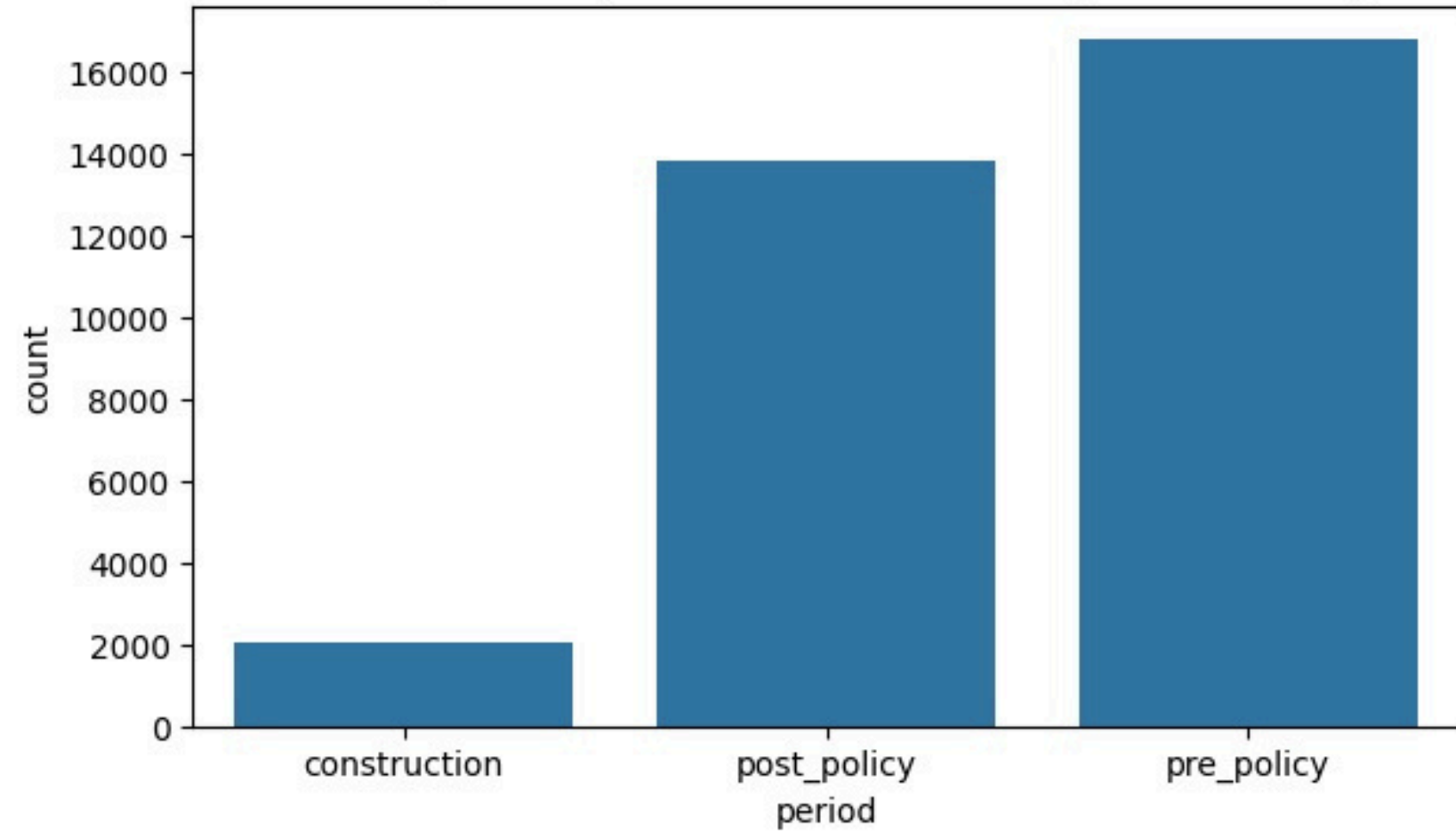
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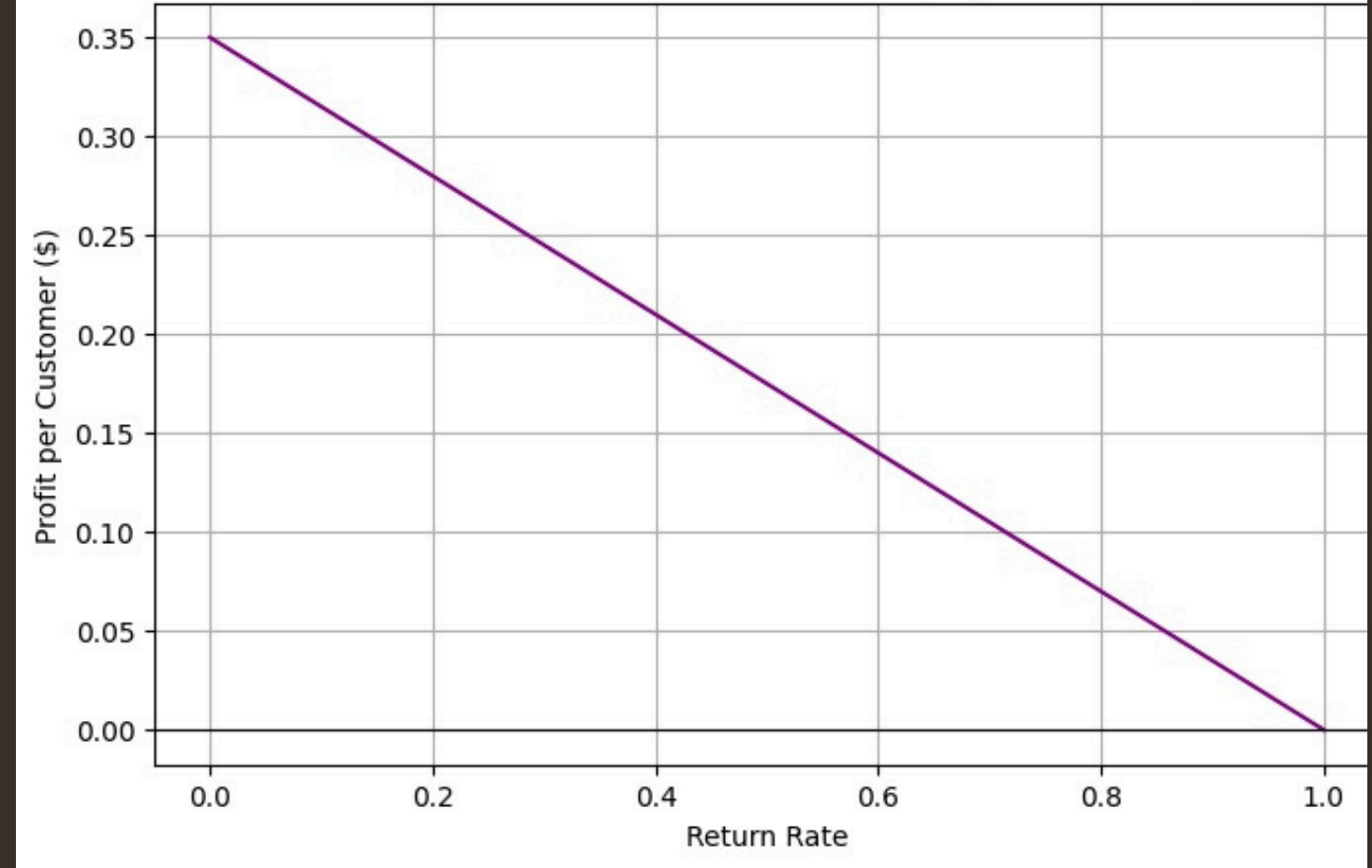
DATA FOR 2 CUPS: === MODEL COEFFICIENTS ===
{'SURCHARGE_APPLIED': NP.FLOAT64(-6.325386750140739),
'HOUR': NP.FLOAT64(0.047072845246706525),
'IS_HOT': NP.FLOAT64(0.992207019793047),
'IS_COFFEE': NP.FLOAT64(-0.030506860555731524)}
INTERCEPT: -1.20892284807263

=== PREDICTIVE SUMMARY ===
SCENARIO PREDICTED AVG REUSABLE CUP PROBABILITY
0
CURRENT POLICY (SURCHARGE)
0.435539
1
DEPOSIT-REFUND
0.605984

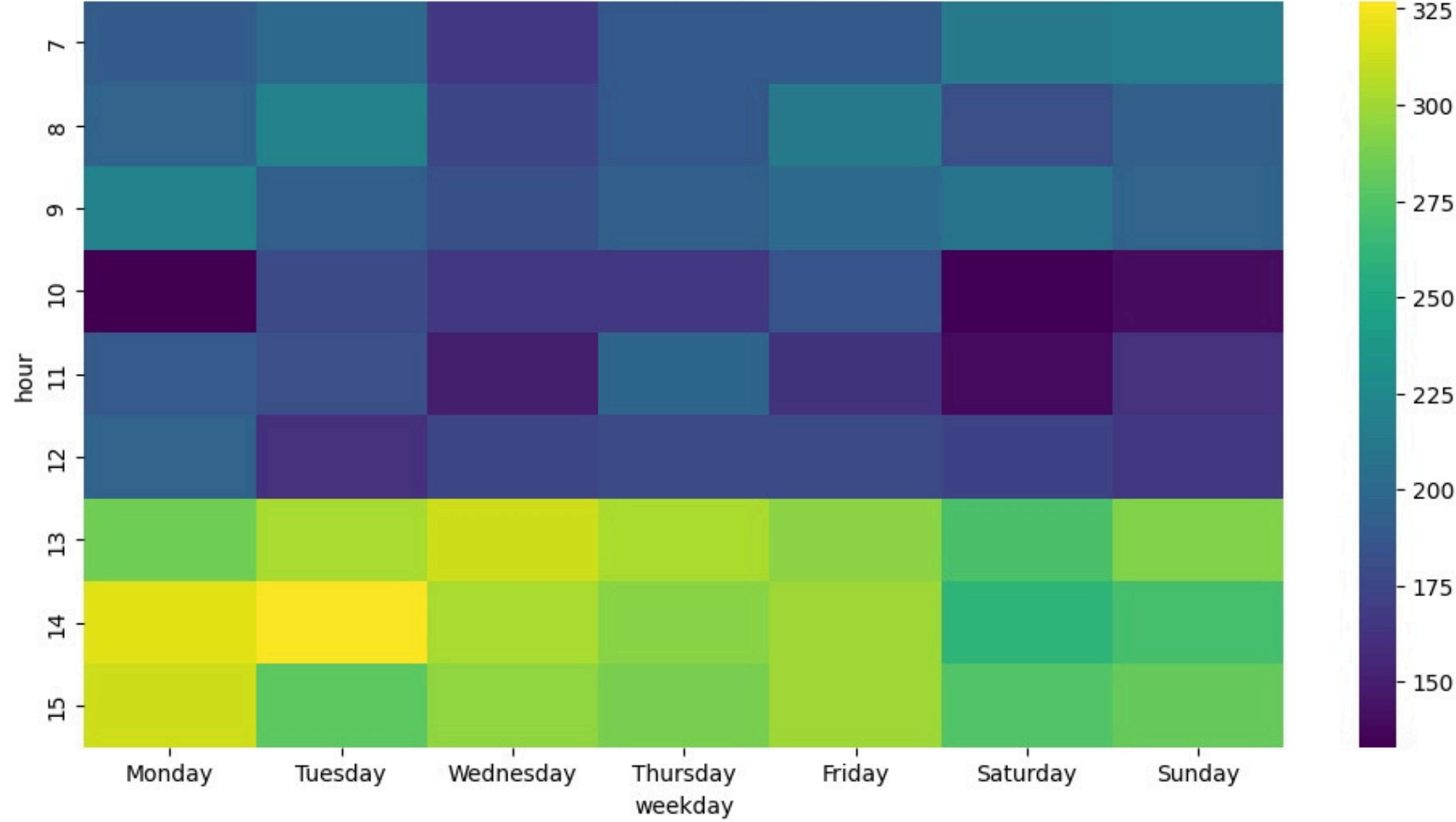
Traffic by Period (Construction → Pre-Policy → Post-Policy)



Profit per Customer vs Return Rate (Deposit-Refund)



Traffic Heatmap — post_policy



Traffic Heatmap — pre_policy

